

1 Quiz question review

Quiz 0

Assume that $y_n = \beta^*_0 + x_n\beta^*_1 + \varepsilon_n$ for scalar x_n and some fixed β^*_0 and β^*_1 . Assume that

- The residuals ε_n are IID with $\mathbb{E}[\varepsilon_n] = 0$ and $\text{Var}(\varepsilon_n) = \sigma^2$.
- The regressors x_n are IID with $\mathbb{E}[x_n] = \mu$ and $\text{Var}(x_n) = \nu^2$.
- The residuals are independent of the regressors.

You should be able to analyze quantities involving x_n , ε_n , and y_n in terms of the fixed quantities β^*_0 , β^*_1 , μ , σ , and ν .

- Be able to compute all the conditional variances and expectations you can think of
- Be able to apply the LLN and CLT, including recognizing when the CLT does not converge
- Be able to apply the LLN to random matrices and their inverses, including recognizing when the matrix is or is not invertible.

In addition, you should be able to

- Write out the $\mathbf{X}$, $\mathbf{Y}$, and $\boldsymbol{\varepsilon}$ matrices for simple linear regressions
- Write out the OLS estimator $\hat{\boldsymbol{\beta}}$ in terms of these quantities
- Be able to use the LLN and CLT to evaluate the limits of

$$\begin{aligned} & - \frac{1}{N} \mathbf{X}^\top \mathbf{X} \\ & - \frac{1}{N} \mathbf{X}^\top \mathbf{Y} \\ & - \frac{1}{\sqrt{N}} \mathbf{X}^\top \boldsymbol{\varepsilon} \end{aligned}$$

- Compute $\mathbb{E}[\hat{\boldsymbol{\beta}}|\mathbf{X}]$ and $\text{Cov}(\hat{\boldsymbol{\beta}}|\mathbf{X})$ in terms of $\text{Cov}(\boldsymbol{\varepsilon})$, and plug in for particular values of $\text{Cov}(\boldsymbol{\varepsilon})$.

In addition, you should be able to do all of the above for the special case when there is no x_n and there is only a constant.

Quiz 1

Suppose we observe data points of the form (y_n, s_n, a_n) , where a_n is a categorical variable and s_n is either a continuous variable or another categorical variable.

You should be able to

- Write the values of the $\mathbf{X}$ matrix for a one-hot encoding of a_n
- Write the values of the $\mathbf{X}$ matrix for interactions between s_n and a_n

- Understand the meaning of interactions (fitting a different slope for each level)
- Write the regression involving the one-hot encoding with and without a constant term, and understand how the interpretation of the coefficients change

Additionally, for all of these combinations, you should be able to do everything from Quiz 0.

Quiz 2

As a starting point, you should be able to do everything for Quiz 0 with a generic vector $\mathbf{x}$ rather than a set of one or two scalars.

You should be familiar with our four assumptions:

- Normality
- Homoskedasticity
- Heteroskedasticity
- Machine learning (IID)

I'll give you what these assumptions are, but you need to be able to understand and apply the definitions.

When we assume that $y_n = \beta^* \mathbf{x}_n + \varepsilon_n$, for each of our four assumptions, with enough hints you should be able to do everything from quiz 0, now in terms of β^* .

For testing, you should be able to analyze the t-statistic for a null of the form $\beta_k^* = 0$ for some component k .

- Know what the distribution of the t-statistic is under the null for normality (all you need is to understand its properties, you don't need to memorize anything)
- Know why you would use an F-statistic, and what its distribution is under the normal assumption (all you need is to understand its properties, you don't need to memorize anything)
- Know the distribution of the t-statistic for large N under non-normal assumptions
- Relatedly, understand how to compute the sandwich covariance and how it is different from the standard covariance
- Be able to compute the level of a given test by applying the above results under the null
- Be able to compute the power of a test against a particular alternative by plugging in an alternative β^*

The question will take the form of undertaking one or more of these tasks under a particular tractable choice of $\mathbf{x}$.

Quiz 3

This question will be about a one-dimensional example $y \sim z$ of a machine learning problem, focusing on squared loss.

You should be able to

- Write out the $\mathbf{X}$ matrix for a zeroth order spline basis of z
- Be able to plot the basis functions
- Understand intuitively what would change with higher-order splines (no need to know the exact formulas)
- Understand the relationship between the zeroth order spline, partitions of the domain of z , and one-hot encodings
- Know what the fitted value $\hat{y}$ is in each partition and be able to plot them (note: this is a generalization of quiz 1)
- Understand intuitively the costs and benefits of having more or fewer partitions

Additionally, you should be able to do all of Quiz 0 for ridge regression. You should also have an intuitive understanding of the costs and benefits of having a large or small ridge parameter.

Quiz 4

This question will be about omitted variable bias in a problem with two regressors which may be correlated with one another. You will need the tools from Quiz 0 for two scalar-valued regressors.

You should be able to

- Relate assumptions about real-world relationships between the regressors to statistical assumptions about their correlation, and use this to derive the limiting behavior of the components of the OLS estimator
- Relate assumptions about real-world causal relationships between the regressors and the values of coefficients in an assumed true data generating process
- Under a misspecified regression, derive the limiting value of the OLS regression coefficient, especially to show when and why it is an incorrect estimator of the assumed true coefficient
- Relate the real-world consequences of omitted variable bias to the mis-estimation of the OLS coefficients under misspecification.

2 Multiple choice questions

Below, you will find the multiple choice questions from the quizzes, but not the candidate responses.

I encourage you to try to think of correct (and incorrect) answers for yourself as a way of preparing for the final exam!

Quiz 0

1. If a square, symmetric matrix $\mathbf{A}$ is invertible, then...
2. Let $\mathbf{X}$ denote a full-rank $N \times P$ matrix with $N > P$. Then...
3. Suppose that $\mathbf{u}$ and $\mathbf{v}$ are orthogonal N -vectors with $N \geq 2$. Then...
4. Let $\mathbf{X}$ denote an $N \times P$ matrix, with $P < N$, $\boldsymbol{\beta}$ a nonzero P -vector, and $\mathbf{Y}$ an N -vector. Which of these matrix expressions is **not** well-defined?
5. Suppose that x_n are IID Gaussian with mean $\mathbb{E}[x_n] = 2$ and $\text{Var}(x_n) = 9$. Which of the following expressions diverges (goes to positive or negative infinity) as $N \rightarrow \infty$?

Quiz 1

1. Suppose that $\mathbf{x}_n = \mathbf{A}\mathbf{z}_n$ for all n , where $\mathbf{A}$ is invertible but not symmetric. For the regressions $y \sim \boldsymbol{\beta}^\top \mathbf{x}$ and $y \sim \boldsymbol{\gamma}^\top \mathbf{z}$, it is necessarily true that...
2. Which of the following is *never* a well justified reason to choose $\hat{\boldsymbol{\beta}}$ by minimizing the mean of squared errors $\frac{1}{N} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \boldsymbol{\beta})^2$?
3. Let $\mathbf{P}_X = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ denote the projection matrix onto the columns of $\mathbf{X}$, where $\mathbf{X}$ is full rank. Which of the following expressions is false?
4. Suppose that x_{n1} and x_{n2} are mean zero and very highly (but not perfectly) positively correlated, and we regress $y_n \sim \beta_1 x_{n1} + \beta_2 x_{n2}$. Then...
5. Let $\mathbf{S}_1, \dots, \mathbf{S}_K$ denote a partition of the space of regressors, so that for each n , $\mathbf{x}_n \in \mathbf{S}_k$ for exactly one k . Let $z_{nk} = \mathbb{I}(\mathbf{x}_n \in \mathbf{S}_k)$, and consider regressing $y \sim \beta_1 z_{n1} + \dots + \beta_K z_{nK}$, without a constant. Assume that each $\mathbf{S}_k$ has at least one observation in it. Then, conditionally on the regressors...

Quiz 2

For the multiple choice questions, recall our four classes of assumptions:

- (A) The normal assumption: $y_n | \mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\beta}^{*\top} \mathbf{x}_n, \sigma^2)$.
- (B) The homoskedastic assumption: $\varepsilon_n = y_n - \boldsymbol{\beta}^{*\top} \mathbf{x}_n$, with $\mathbb{E}[\varepsilon_n | \mathbf{x}_n] = 0$, and $\text{Var}(\varepsilon_n | \mathbf{x}_n) = \sigma^2$.

- (C) The heteroskedastic assumption: Like (B), but $\text{Var}(\epsilon_n | \mathbf{x}_n) = \sigma_n^2$, where σ_n is some function of $\mathbf{x}_n$ that is different for different $\mathbf{x}_n$.
- (D) The machine learning assumption: $(\mathbf{x}_n, y_n)$ are IID.

In each assumption assume that the pairs $(\mathbf{x}_n, y_n)$ are IID and that $\mathbf{X}$ is full-rank, and that all necessary conditions hold for the application of the CLT and LLN as in the lecture notes.

Recall that $\hat{\sigma}^2 = \frac{1}{N-P} \sum_{n=1}^N (y_n - \hat{\beta}^\top \mathbf{x}_n)^2$.

1. Under the normal assumption (A), under the null, the standard t-statistic based on $\hat{\sigma}$...
2. Under the homoskedastic assumption (B), under the null, the standard t-statistic based on $\hat{\sigma}$...
3. Under the heteroskedastic assumption (C), under the null, the standard t-statistic based on the sandwich covariance...
4. Under the heteroskedastic assumption (C), inference that is incorrectly based on the homoskedastic assumption (B) will, in general...
5. Let $\beta^* := \mathbb{E}[\mathbf{x}\mathbf{x}^\top]^{-1} \mathbb{E}[\mathbf{x}y]$. Under the machine learning assumption (D),

Quiz 3

1. In a machine learning problem with ridge regression, it is typically best practice to select the ridge parameter using...
2. When running ridge regression, as you *decrease* the ridge penalty parameter...
3. For machine learning, a good reason to use a highly expressive spline basis is...
4. For machine learning, you don't want your spine basis to be too expressive because...
5. In Bayesian analysis, the ridge parameter corresponds to...

Quiz 4

1. Suppose that you want to use “academic ability” as a regressor, but only have access to test results that are unbiased but suffer from random variation. Regressing on these noisy test results rather than a higher quality measure of academic ability...
2. Suppose that you have a regression problem with one very extreme outlying regressor x_n . Then...
3. Suppose that you have a regression problem with one very extreme outlying response y_n , but typical $\mathbf{x}_n$. Then...
4. When there are extreme outliers in a dataset...
5. In linear regression, omitted variable bias occurs...